



School of Future Tech

Case Study Report

on

Retail Stockout Alert Service ReStock: Retail Inventory Management

Digital Transformation: Inventory
Go-to-Market & Customer Operations

by

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Portfolio: yashhh710.github.io/restock-inventory-management
Source code: github.com/Yashhh710/restock-inventory-management

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1. Introduction to the Case Study

Small retailers such as kirana stores, mini markets, convenience stores and small pharmacies depend on a handful of fast-moving products. When one of these runs out, the shopkeeper usually finds out only when a customer asks for it. The sale is lost, and the customer may shop elsewhere next time.

This case study, **Retail Stockout Alert Service**, is about helping such a retailer see a stockout coming before it happens. The product built for it is **ReStock**: a simple inventory-management concept that links four everyday activities, **stock recording, sales tracking, low-stock detection and replenishment**.

The work has three parts: a product portfolio website that explains the idea, an interactive browser-based prototype with twelve working pages, and business-model material (value proposition, customer journey, go-to-market plan, SWOT and risks). As an extra check on the idea, the project was also reviewed by nine AI assistants, and their feedback is analysed in Section 5.5.

2. Problem Statement / Case Background (Abstract)

Case Details

Item	Detail
Case	Retail Stockout Alert Service
Topic	Digital Transformation: Inventory
Product	ReStock: Retail Inventory Management
Course	Go-to-Market & Customer Operations
Programme	B.Tech Computer Science Engineering, ITM Skills University
Sprint	Semester III, Sprint I
Live portfolio	yashhh710.github.io/restock-inventory-management
Prototype	yashhh710.github.io/restock-inventory-management/main-product's-prototype/dashboard.html
Source code	github.com/Yashhh710/restock-inventory-management

Problem Statement

Small retailers often discover stockouts only when customers ask for products. ReStock addresses this with a simple digital workflow in which a shopkeeper records opening stock and sales for selected fast-moving products. The system flags products that reach or fall below their reorder levels and gives actionable replenishment information. The academic pilot scenario is built around a store tracking 80 products; the browser prototype uses a smaller seeded catalogue of 10 products to demonstrate the workflow.

Objectives

- Identify common stockout pain points for small retailers.

- Simplify daily stock recording and sales tracking.
- Provide clear low-stock and out-of-stock alerts.
- Connect inventory information with purchasing and replenishment.
- Provide reorder suggestions using stock level and sales velocity.
- Give the retailer one simple operational dashboard.
- Create a practical pilot-to-scale business model.

Abstract

ReStock is a static web product made of a portfolio site and a vanilla-JavaScript prototype. Shopkeepers can add products, record sales and purchases, adjust stock, view alerts, create reorder requests, manage suppliers and read stock history and reports. A product is "Low Stock" when current stock is at or below its reorder level and "Out of Stock" when it is zero. A suggested reorder quantity is calculated from daily sales and the reorder level. An "AI Insights" page turns the same data into recommendations such as estimated days of stock remaining; the prototype labels these as demo insights, not production-grade AI. Beyond the prototype, the report develops the customer profile, value proposition, journey, supply chain, business model, 90-day pilot plan, SWOT and risk analysis. Nine AI assistants reviewed the idea. All nine judged the problem real, and all nine felt the AI feature was the weakest part. This feedback is used to set the product roadmap.

Target Customers and Pain Points

Primary customer: small retail shopkeepers and store operators managing fast-moving products who rely on manual stock checks, notebooks, spreadsheets or memory.

Secondary users: store assistants, kirana, convenience and mini-market operators, small pharmacies, and inventory or purchasing staff.

- Stockouts are discovered too late, and there is no single view of current stock.
- Low-stock products are hard to monitor manually, and sales information is separate from purchasing decisions.
- Reordering is reactive rather than planned, and supplier information is scattered.
- Stock movements are hard to trace, and manual calculations lead to inconsistent reorder decisions.

Customer Profile: The Small Retail Shopkeeper

The shopkeeper needs to keep fast-moving products available without building up unnecessary stock. The current workflow is: receive stock from suppliers, place it on shelves, record or remember sales, check stock manually, notice low stock during routine inspection, contact a supplier, and reorder. The core problem is the gap between sales happening and the shopkeeper noticing that inventory has become low. A shopkeeper using ReStock should be able to answer quickly: What is in stock? What is low? What is out of stock? What sold today? What should I reorder? Which supplier should get the order?

3. Case Study Design

3.1 Product Vision and Design Principles

The vision is a single idea: help a retailer know what is running low before the customer discovers the stockout. The workflow is Stock Recording, then Sales Tracking, then Low-Stock Detection, then Replenishment. Seven principles guide the design: simple, action-oriented, inventory-first, data-connected, scalable, transparent (recommendations show the data behind them) and retail-focused.

3.2 Value Proposition Canvas

Customer Jobs	Customer Pains	Customer Gains	How ReStock responds
Keep products available; record sales; monitor stock; reorder; talk to suppliers; avoid overstock; know which products move fast.	Unexpected stockouts; manual counting; missed reorder points; no view of sales velocity; time spent checking several records; no simple dashboard.	Earlier stockout visibility; faster reorder decisions; one place for product data; better supplier coordination; less manual monitoring; consistent reorder decisions.	Pain relievers: low-stock and out-of-stock alerts, reorder suggestions, stock history, supplier records. Gain creators: dashboard, sales and stock together, one-click reorder workflow, reports.

3.3 Customer Journey

Stage	Customer action	Need	ReStock touchpoint
Discover	Learns about inventory problems	Understand the value	Product portfolio
Evaluate	Explores the workflow	Confirm suitability	Live demo / prototype
Onboard	Adds store information	Start quickly	Settings
Setup	Adds products and reorder levels	Establish the catalogue	Products
Operate	Records sales	Keep inventory current	Sales
Monitor	Checks stock	Detect risk	Dashboard
Alert	Sees low-stock information	Know what needs action	Alerts
Reorder	Creates a replenishment request	Restore stock	Reorder Requests
Purchase	Records the supplier purchase	Track incoming stock	Purchases
Review	Checks performance	Understand operations	Reports
Improve	Reviews recommendations	Improve replenishment	AI Insights

3.4 End-to-End Operating Flow

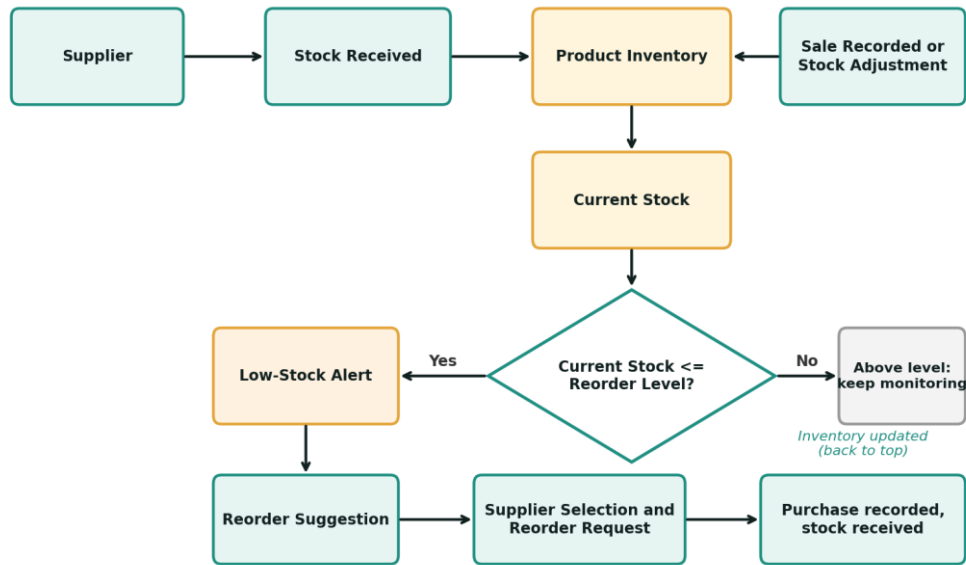


Figure 1: ReStock end-to-end operating flow

Stock enters from suppliers; sales and adjustments take it out. After every change, the current stock is compared with the product's reorder level. If it is at or below the level, an alert is raised, a reorder quantity is suggested, a supplier is selected and a reorder request is created. When the purchase arrives, stock is received and inventory is updated.

3.5 Supply Chain

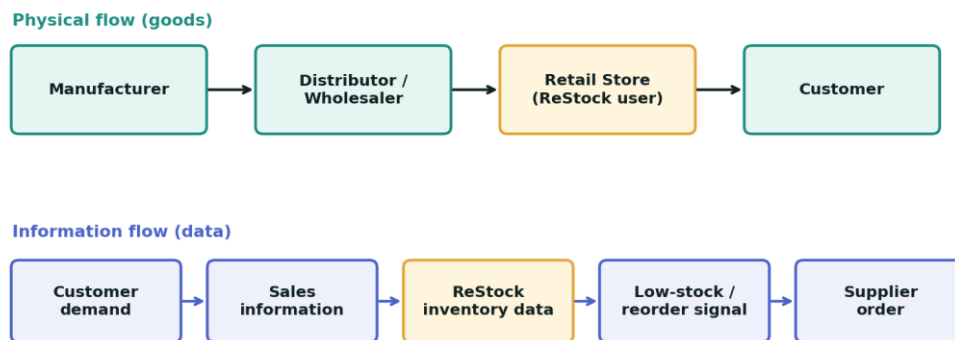


Figure 2: Physical and information flow in the retail supply chain

Participant	Responsibility
Manufacturer	Produces goods
Distributor / wholesaler	Supplies products to retailers
Retailer / shopkeeper	Maintains inventory, serves customers, records sales and stock movements
Supplier	Fulfils replenishment orders
ReStock	Organises inventory information and operational alerts
Customer	Purchases products and creates demand

3.6 Business Model

Revenue streams. SaaS subscriptions (Starter, Business, Multi-store), premium analytics as an add-on, multi-store plans (central reporting, user management), and possible future supplier partnerships. **Cost structure.** Product development, hosting, database and notification infrastructure, maintenance, customer support, marketing, retailer acquisition and onboarding.

The portfolio shows proposed pricing plans. These are illustrative; no payment system exists.

Plan	Price	Limit	Included
Starter	Free	Up to 30 products	Basic inventory tracking, low-stock alerts, daily sales recording, basic dashboard
Retail	Rs 299 / month	Up to 500 products	Automatic stock alerts, sales history, reorder management, supplier tracking, inventory reports
Enterprise	Custom quote	Larger needs	Custom plan on request

Business Model Canvas:

Block	ReStock
Customer segments	Small retailers, kirana stores, convenience stores, small pharmacies
Value proposition	Simple stock monitoring, low-stock alerts and a replenishment workflow
Channels	Direct sales, retailer pilots, distributor partnerships, digital marketing
Customer relationships	Guided onboarding, support, self-service dashboard
Revenue streams	SaaS subscriptions, premium analytics, multi-store plans, future supplier partnerships
Key resources	Product software, inventory logic, customer data, hosting infrastructure
Key activities	Product development, onboarding, support, analytics, retailer acquisition
Key partners	Distributors, wholesalers, retail associations, technology providers
Cost structure	Development, hosting, support, marketing, infrastructure and operations

3.7 Go-to-Market Strategy

The initial pilot targets small kirana stores, convenience stores, mini markets, small pharmacies and independent retailers with fast-moving products. Two acquisition methods are planned.

- **Local retailer pilot:** identify retailers, demonstrate the prototype, configure the product catalogue, train the shopkeeper, run a 30-day pilot, collect usage data, improve the product.

- **Distributor partnerships:** work with distributors who already serve many small retailers, offer simple onboarding and a referral mechanism, and use pilot results to show operational outcomes.

3.8 90-Day Pilot-to-Scale Plan

Phase	Objectives	Success criteria
Days 1-30: Pilot setup	Recruit pilot retailers; configure catalogues; define reorder levels; train store operators; capture baseline inventory metrics.	Stores onboarded; catalogues configured; operators trained; daily stock and sales workflow understood; baseline stockout and reorder metrics captured.
Days 31-60: Usage and optimisation	Monitor actual usage; find workflow friction; improve alerts and reorder recommendations; collect retailer feedback.	Consistent weekly usage; alerts reviewed regularly; reorder workflow used; feedback documented; improvements prioritised.
Days 61-90: Scale preparation	Standardise onboarding; package pricing; prepare sales material; document support; expand to more stores.	Repeatable onboarding; defined plans; measurable pilot outcomes; documented customer feedback; scale-ready process.

3.9 SWOT Analysis

Strengths	Weaknesses
Simple inventory workflow; clear focus on stockout prevention; easy-to-understand reorder-level concept; dashboard centralises information; prototype shows several workflows; designed around small-retailer needs.	Prototype uses seeded demo data stored only in the browser; no production database; AI Insights are demonstrative; supplier integrations are not live; accuracy depends on consistent data entry.
Opportunities	Threats
Multi-store inventory; supplier and distributor integrations; barcode scanning; mobile-first workflows; automated notifications; demand forecasting; purchase-order automation; POS integration; retail analytics.	Existing inventory apps; retailer resistance to changing manual habits; poor data-entry discipline; connectivity limits; wrong reorder thresholds; supplier delays; price changes; customer acquisition cost.

3.10 Risk Analysis and Mitigation

Risk	Potential effect	Mitigation
Incorrect stock data	Wrong alerts	Validation and stock-history records
Missed sales entries	Inventory becomes inaccurate	Simple sales-entry workflow and reminders
Incorrect reorder level	Ordering too early or too late	Configurable reorder levels
Supplier delay	Stock stays unavailable	Track supplier lead time; keep safety stock
User adoption issues	Low product usage	Simple onboarding and short training
Poor connectivity	Workflow interruptions	Local persistence; resilient web or mobile design in production

Risk	Potential effect	Mitigation
Excessive alerts	Alert fatigue	Prioritise urgent products; configurable notifications
Incorrect AI recommendation	Poor reorder decision	Explain recommendations; allow manual override
Data security issue	Loss of trust	Authentication, authorisation and secure storage in production
Scaling infrastructure	Performance degradation	Monitoring and scalable hosting

3.11 Product Architecture

The repository is a static portfolio plus a front-end prototype. The portfolio has Home, Products, Services, Blog and Contact pages. The prototype has twelve pages sharing one script (app.js) and one stylesheet (style.css). Supporting folders hold product models (models/) and the AI review material (Ai-rewive/).

4. Methods, Algorithms & Technology Applied

4.1 Stock Status Rule

Every product is in one of three states, decided by a simple rule in the prototype:

```
const stat = p => p.stock === 0 ? "out" : p.stock <= p.reorderLevel ? "low" : "ok";

// out -> "Out of Stock"   low -> "Low Stock"   ok -> "Healthy"
```

4.2 Stock Cover and Suggested Reorder Quantity

Stock cover estimates how many days current stock will last at the average sales rate:

Stock Cover (days) = Current Stock / Average Daily Sales. Example: 8 units at 4 units per day gives 2 days. This assumes the sales rate stays constant.

Suggested order quantity. The prototype uses: Suggested = max(Daily Sales x 5, Reorder Level x 2 - Current Stock, 1). It orders at least five days of sales, or enough to reach twice the reorder level, and never less than one unit. The README also describes a simpler pilot formula (Target Stock - Current Stock) and a production formula that adds safety stock, supplier lead time, seasonality and minimum order quantities.

```
function suggest(p) {
  return Math.max(p.dailySales * 5, p.reorderLevel * 2 - p.stock, 1);
}
```

Applying these rules to the 10 seeded products of the prototype gives the following results:

Product	Stock	Reorder level	Daily sales	Status	Cover (days)	Suggested order
Maggi 70g	8	10	4	Low stock	2.0	20
Coca-Cola 750ml	6	12	5	Low stock	1.2	25
Aashirvaad Atta 5kg	0	8	3	Out of stock	0	16

Product	Stock	Reorder level	Daily sales	Status	Cover (days)	Suggested order
Parle-G Biscuits	7	10	2	Low stock	3.5	13
Tata Salt 1kg	15	10	1	Healthy	15.0	5
Amul Milk 1L	34	20	6	Healthy	5.7	30
Lays Classic 52g	22	12	3	Healthy	7.3	15
Dettol Soap 75g	41	15	1	Healthy	41.0	5
Crocin Advance	28	10	2	Healthy	14.0	10
Surf Excel 1kg	19	8	1	Healthy	19.0	5

Maggi 70g (8 in stock, reorder level 10, 4 sold per day) is Low Stock with 2 days of cover and a suggested order of 20 units. This matches the Reorder Product window in Figure 17. Aashirvaad Atta is out of stock and a suggested 16 units appears in the AI Insights page.

4.3 How the "AI Insight" Works

The AI Insight card finds the product with stock above zero and daily sales above zero that has the lowest cover (stock divided by daily sales). If that cover is 7 days or less, it shows "may need replenishment soon" with current stock, average daily sales and about how many days remain, and offers a Create Reorder button. Otherwise it reports that stock levels look stable. This is a rules-based calculation on demo data, which is why the page is labelled "Demo AI Insight: not production-grade AI". Section 5.5 shows that independent AI reviewers noticed the same thing.

```
const c = S.products.filter(p => p.stock > 0 && p.dailySales > 0)
  .map(p => ({ p, d: p.stock / p.dailySales }))
  .sort((a, b) => a.d - b.d)[0]; // lowest days of cover
if (c && c.d <= 7) { /* "may need replenishment soon" + Create Reorder */ }
else { /* "Stock levels look stable." */ }
```

4.4 Performance Metrics

Metric	Formula
Stockout rate	Stockout events / tracked product-periods
Alert response rate	Alerts acted upon / alerts generated
Reorder completion rate	Completed reorders / created reorders
Average stock cover	Current stock / average daily sales
Inventory availability	Available product-days / total product-days
Active retailers	Retailers using the product during the period

Illustrative pilot arithmetic: if a store tracks 80 products and 12 are below their reorder levels, then low-stock percentage = $12 / 80 \times 100 = 15\%$. This is a snapshot of products below reorder level, not a stockout rate. These figures describe the academic pilot scenario and are not measured results from a real store.

4.5 Data Model and Persistence

A production version would use these logical entities: Store, Product (with Category and Supplier), Sale, Purchase, Reorder Request, Stock Movement and User. The prototype has no database. It keeps everything in the browser's localStorage under three keys: restock_v2 (inventory, sales, purchases, reorders, suppliers, history), restock-settings and restock-theme. Each stock change is also written to a Stock History record with the previous stock, new stock and a reference.

Entity	Main fields
Product	Product ID, name, SKU, category, current stock, reorder level, daily sales, purchase price, selling price, supplier ID
Sale	Sale ID, product ID, quantity, selling price, total, date, user ID
Purchase	Purchase ID, product ID, supplier ID, quantity, purchase price, date, status
Reorder request	Request ID, product ID, supplier ID, suggested quantity, requested quantity, status (Pending, Ordered, Received), created date
Stock movement	Movement ID, product ID, type (Sale, Purchase, Adjustment, Stock Received), quantity, previous stock, new stock, reference, timestamp

4.6 Technology Stack

Part	Technology
Portfolio site	HTML, CSS, JavaScript, Astro-generated static assets, responsive design
Prototype	HTML5, CSS3, vanilla JavaScript, browser localStorage, inline SVG icons and charts, responsive layout with dark and light themes
Deployment	GitHub Pages (static hosting)
Run locally	python3 -m http.server 8000 inside the prototype folder, or VS Code Live Server
Version control	Git and GitHub

5. Implementation Details and Snapshots

5.1 Product Portfolio Website

The portfolio is the public face of the product. It has a hero section ("See Low Stock Before Customers Do"), the problem explained as three pains (missed stockouts, manual tracking, lost sales), three core features (Smart Stock Alerts, Inventory Dashboard, Easy Reordering), a low-stock-to-reorder walkthrough, proposed pricing, an FAQ, and a link to the live demo, the GitHub repository and the report. Sign-in and sign-up forms on the site are marked "Demo only" and are not connected to a backend. The hero badge shows "Reviewed by 9 AI Models" with a 4.1 / 5 combined rating, which comes from the AI review exercise in Section 5.5.

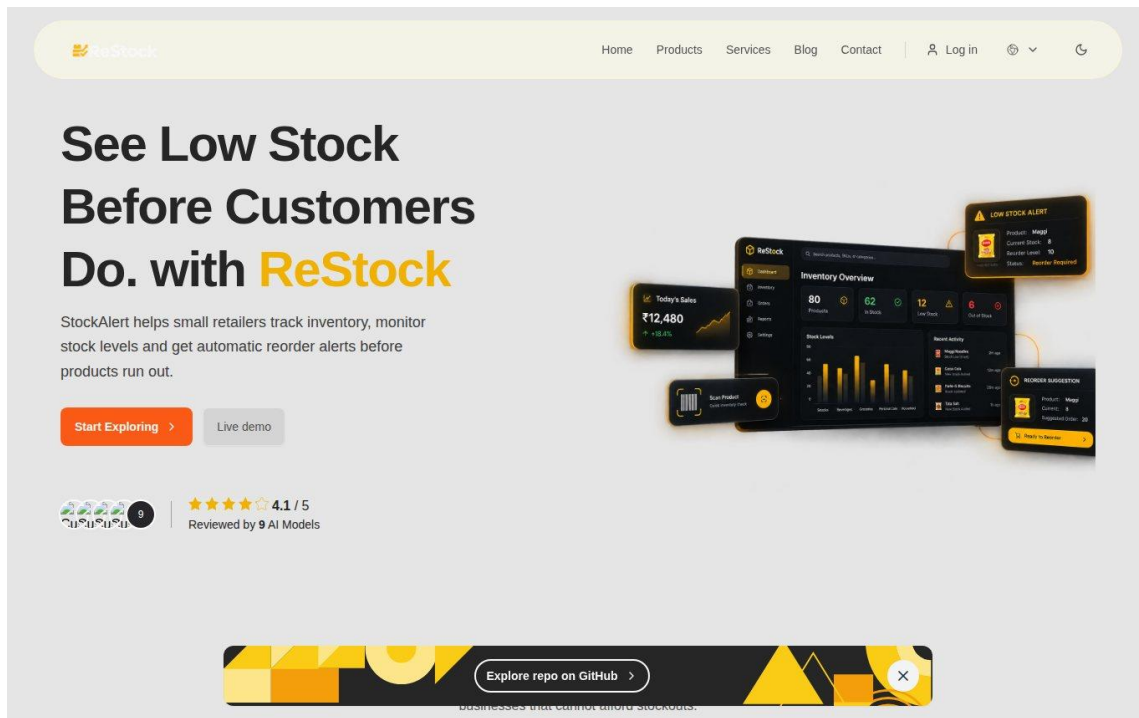


Figure 3: ReStock portfolio home page

5.2 Prototype Modules

The prototype opens on the Dashboard and has twelve pages. Each screenshot below was captured from the running prototype. Note that the dashboard totals (128 products, 12 low stock, 4 out of stock, 2,486 units) combine the 10 seeded products with additional demo placeholder figures that make the store look larger; the 10-product table in Section 4.2 is the real working data.

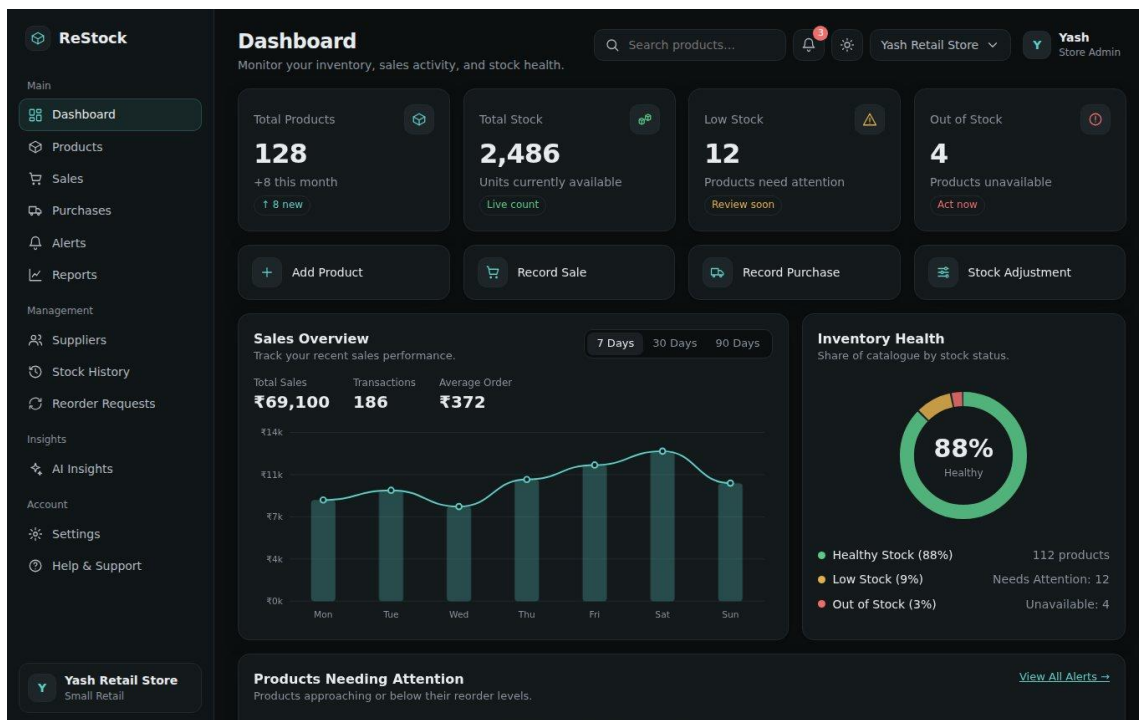


Figure 4: Dashboard with KPIs, sales overview, inventory health and quick actions

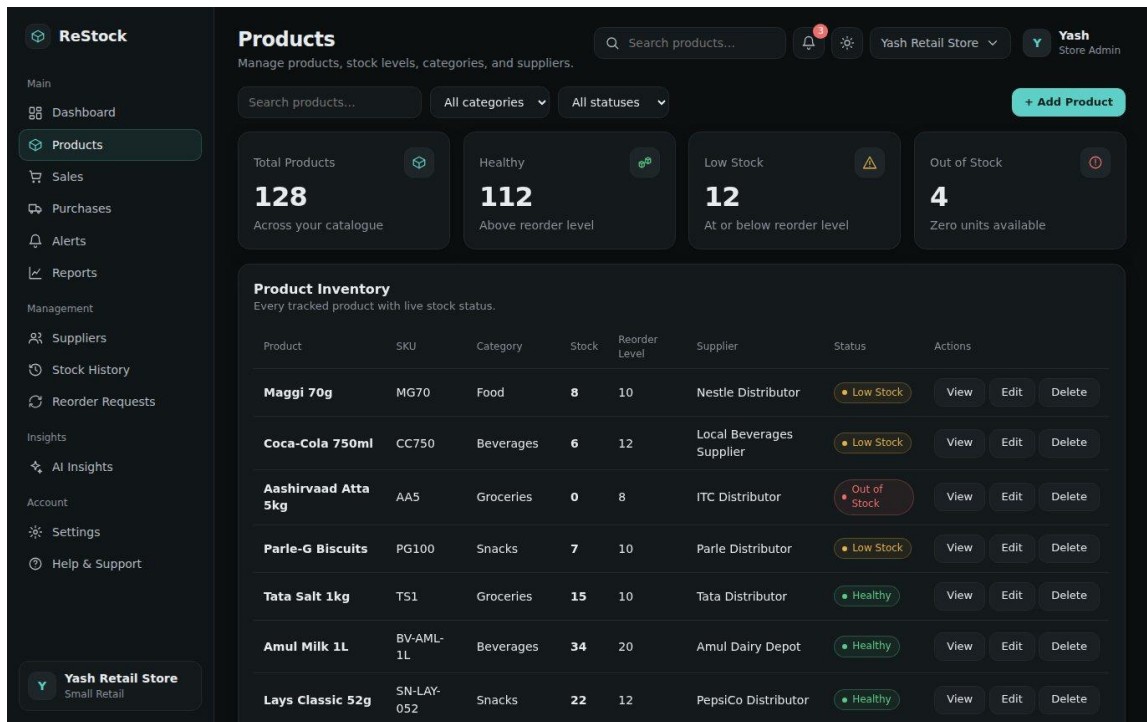


Figure 5: Products page with search, category and status filters, and view / edit / delete actions

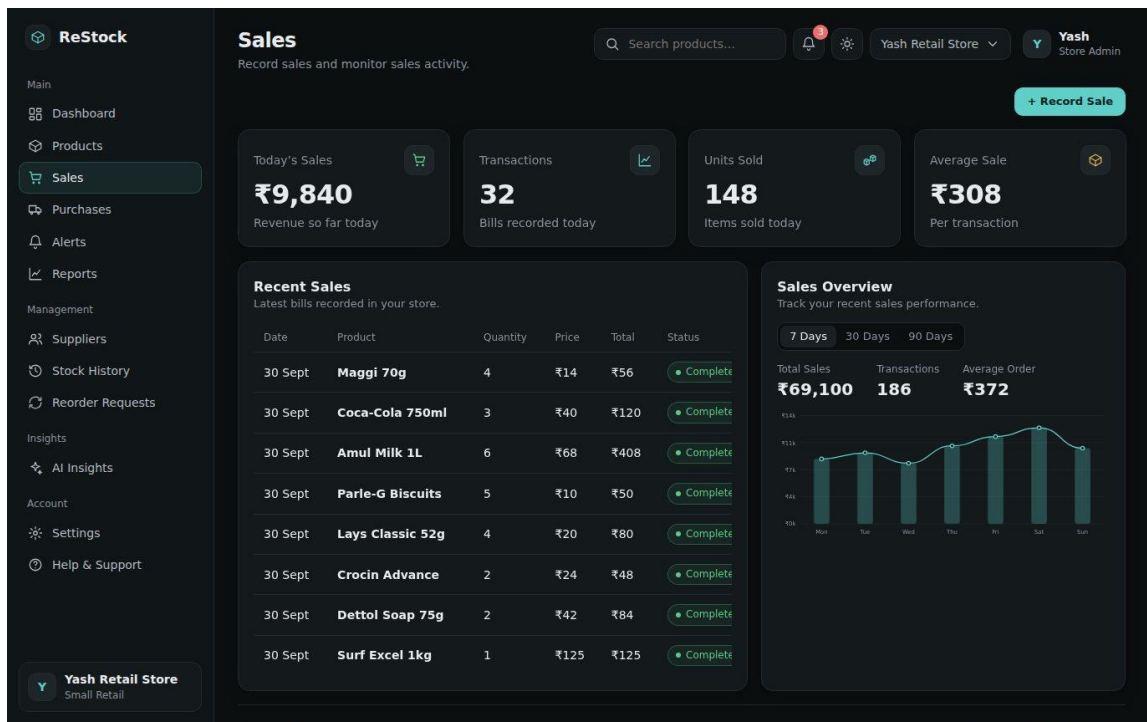


Figure 6: Sales page with recent sales and sales overview

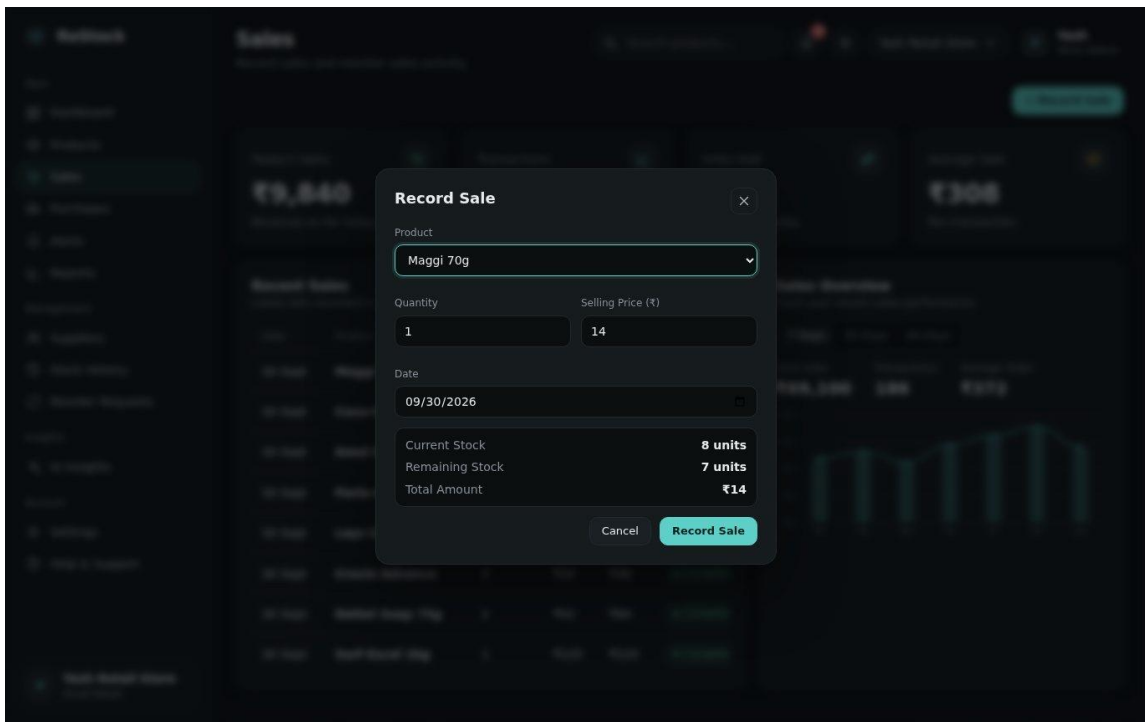


Figure 7: Record Sale window showing current stock, remaining stock and total amount

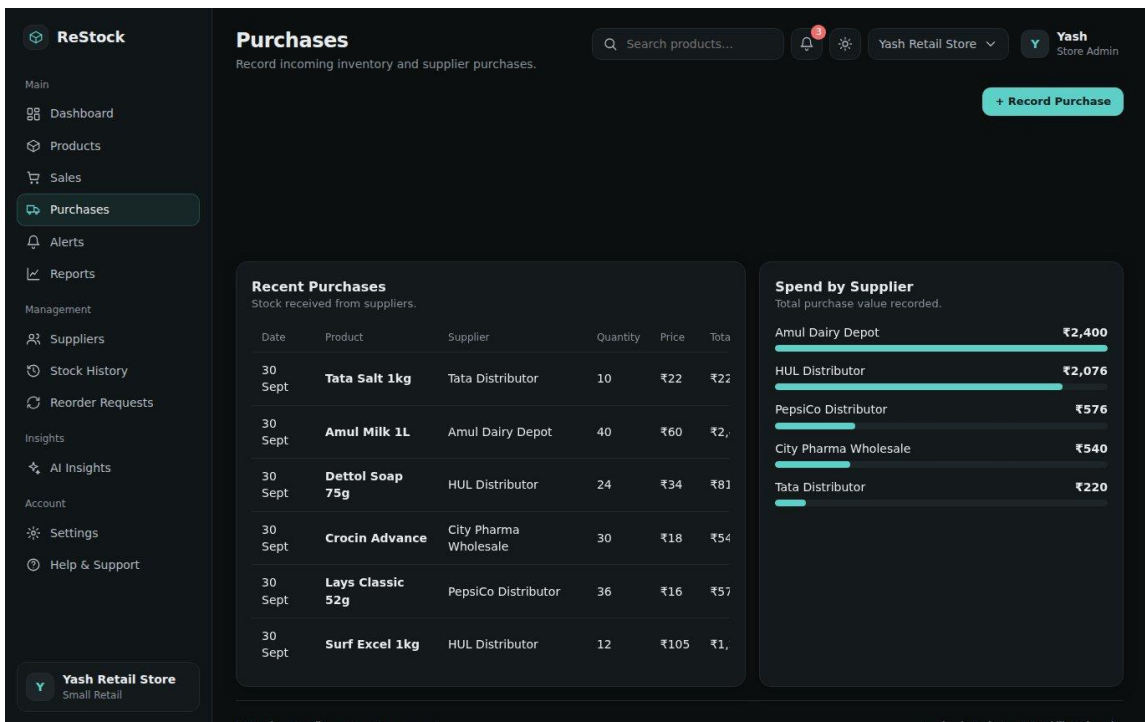


Figure 8: Purchases page with spend by supplier

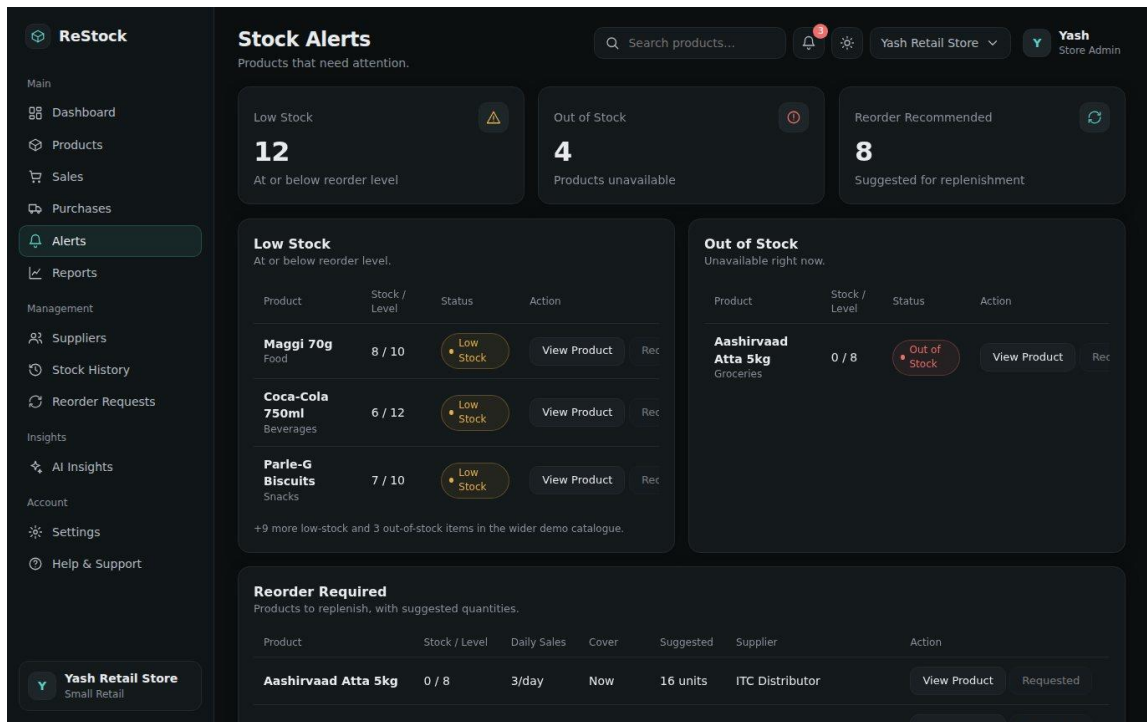


Figure 9: Stock Alerts page: low stock, out of stock and reorder required

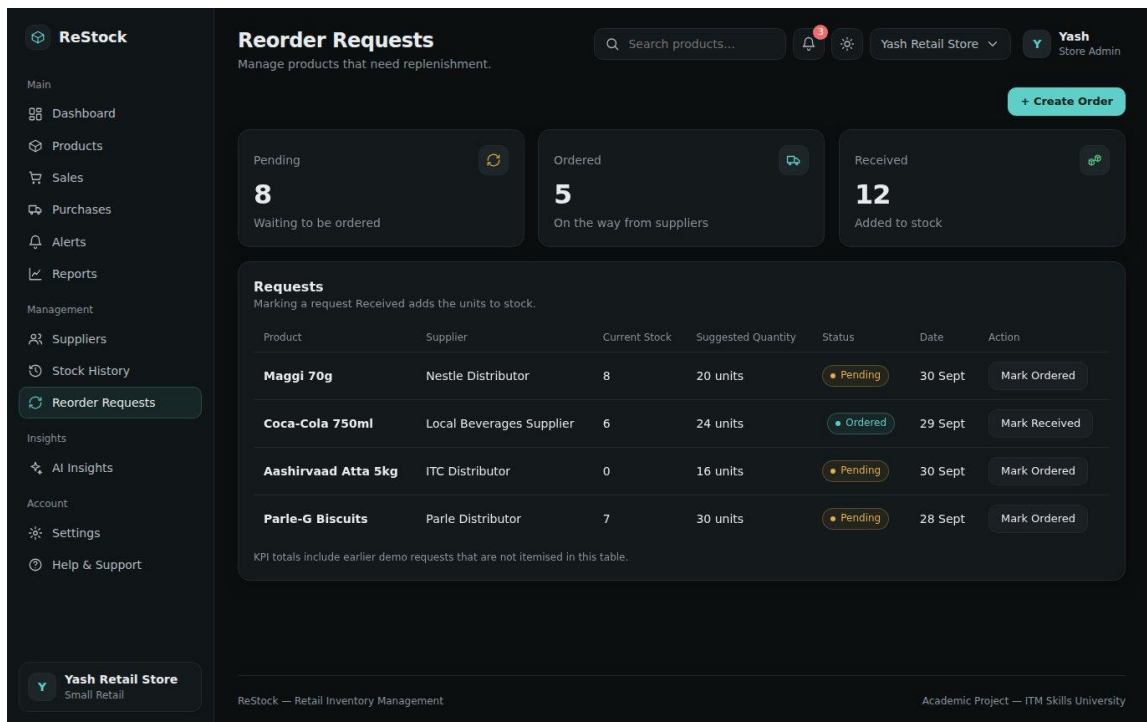


Figure 10: Reorder Requests with Pending, Ordered and Received status

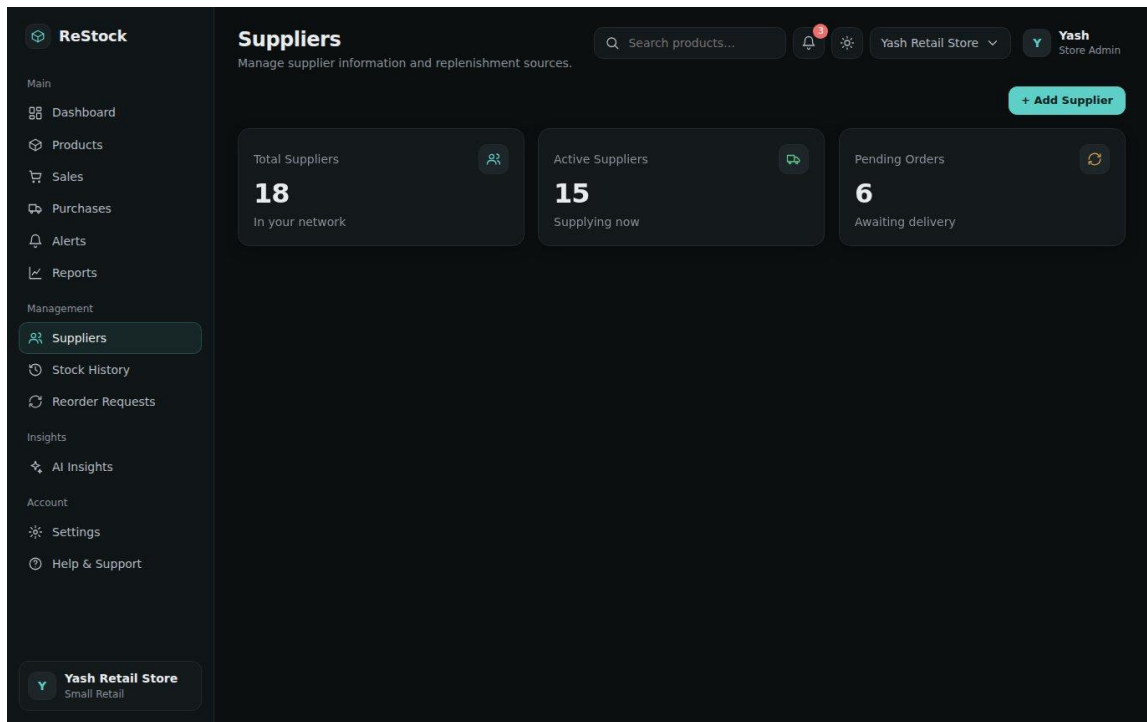


Figure 11: Supplier directory

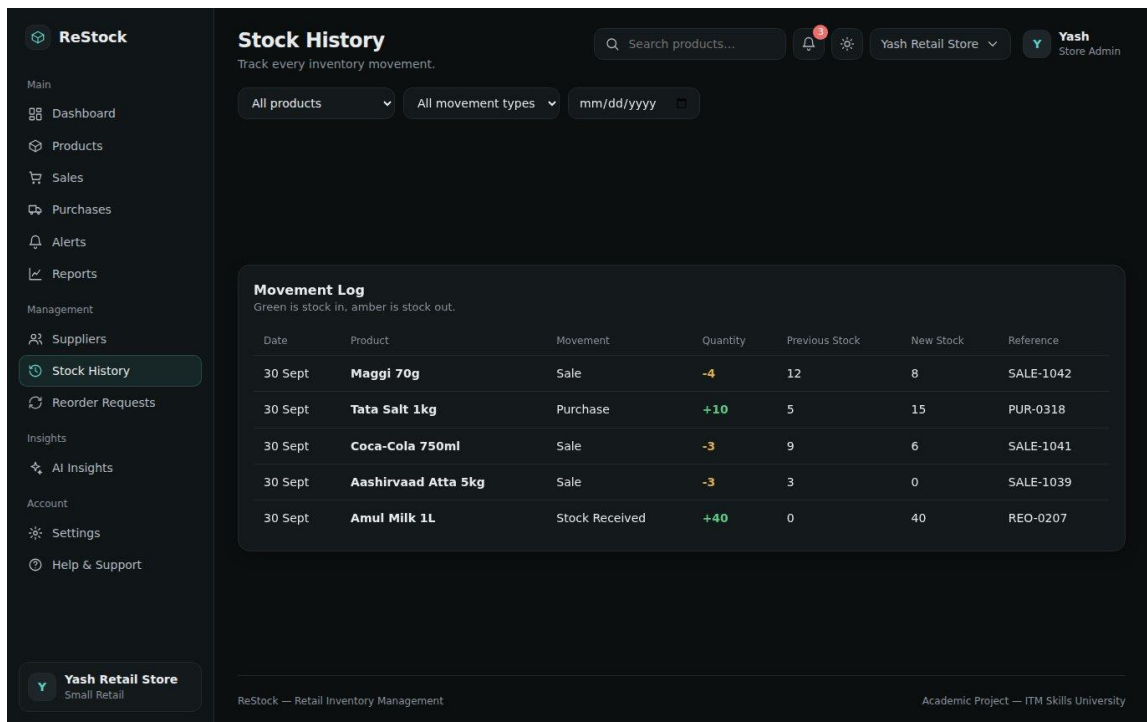


Figure 12: Stock History movement log (date, product, type, quantity, previous and new stock, reference)



Figure 13: Reports and Analytics

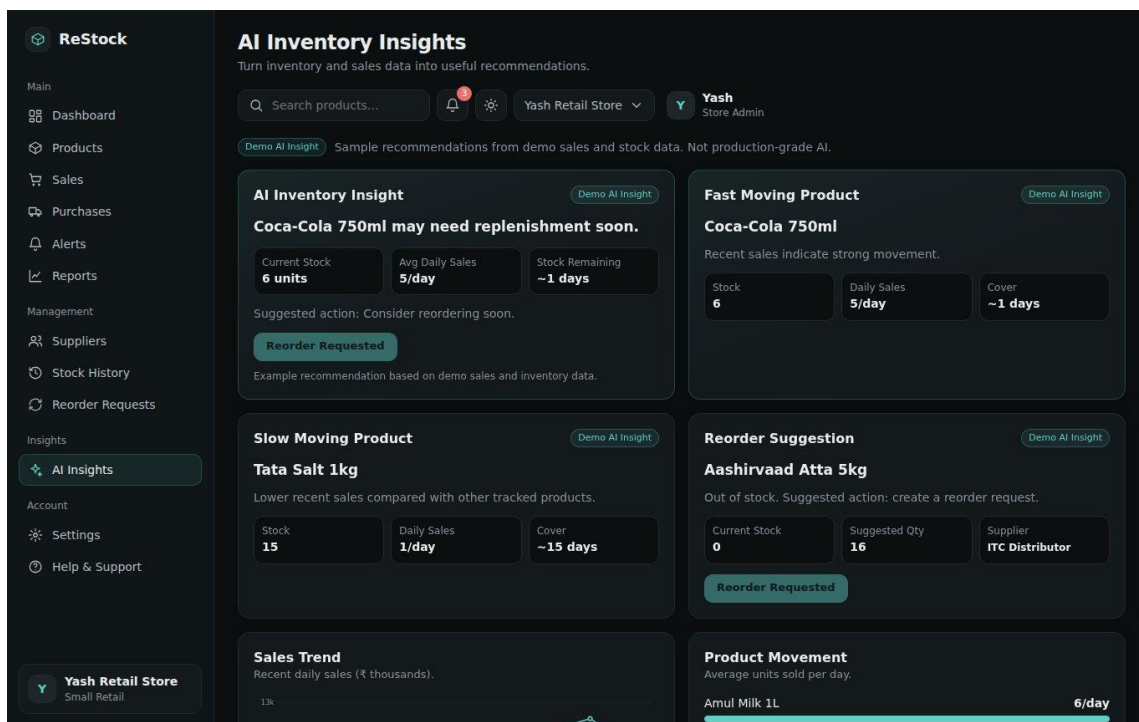


Figure 14: AI Inventory Insights marked as a demo

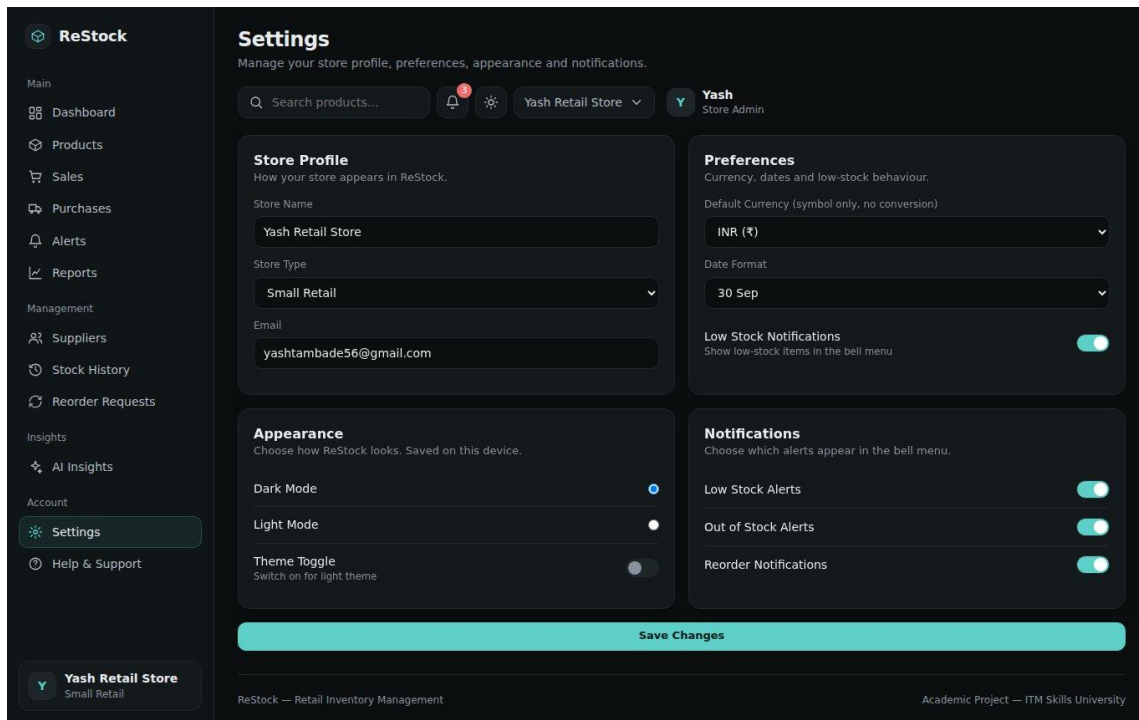


Figure 15: Settings for store profile, preferences, appearance and notifications

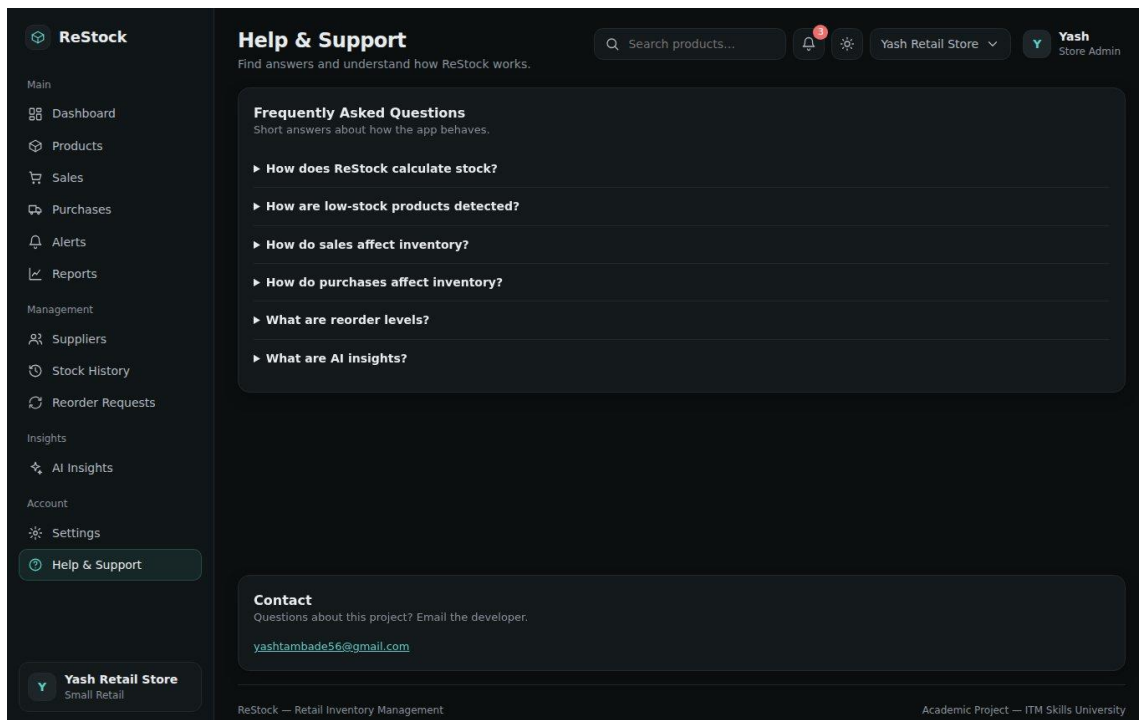


Figure 16: Help and Support with FAQ

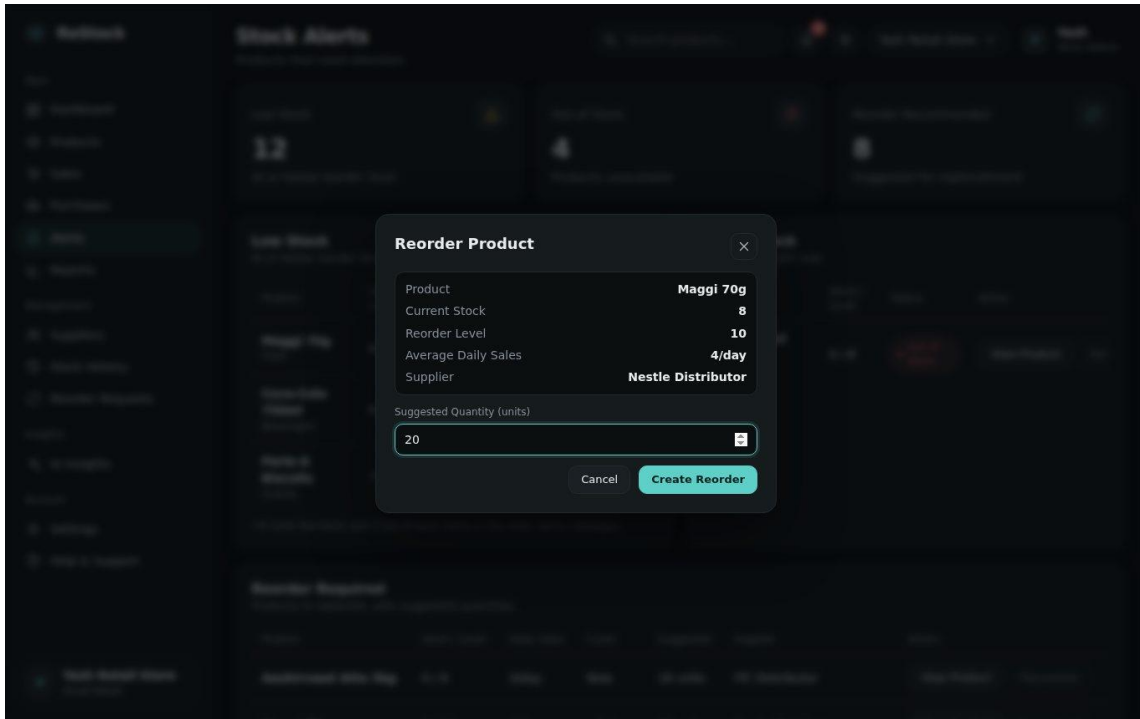


Figure 17: Reorder Product window for Maggi 70g with a suggested quantity of 20

5.3 Key Workflow Logic

Receiving a reorder is what closes the loop: marking a reorder request as Received adds its quantity to the product stock, increases the "stock in" movement count, and records the change in Stock History.

```
function setRQ(id, st) {
  const r = S.reorders.find(x => x.id == id);
  const p = S.products.find(x => x.name === r.product);
  if (st === "Received" && r.status !== "Received" && p) {
    hint = "Stock Received";
    p.stock += r.quantity; S.mv.in += r.quantity;
    log("purchase", `${p.name} x ${r.quantity} received`);
  }
  r.status = st; renderAll();
}
```

Recording a sale is validated so that stock can never go negative: if the quantity is more than the stock on hand, the sale is rejected with a message such as "Only 8 units of Maggi 70g in stock". After a sale, the status is re-evaluated and a new alert is logged if the product has just reached its reorder level or run out.

5.4 Running the Prototype

```
cd "main-product's-prototype"
python3 -m http.server 8000
# open http://localhost:8000/dashboard.html
```

5.5 AI Reviews of the Project Idea

As an independent check of the idea, ReStock was reviewed by nine AI assistants. All of them received the same prompt, which explained the project, the problem, the proposed workflow, the main features and the Maggi example, and said clearly that this is a student

project, not a production-ready product. The prompt asked each assistant to review only the project idea (not the website design) and to answer in a fixed format: one sentence each for Positive, Negative, Suggestion and Overall, with a "Reviewed by" line, under 90 words. The rules told them not to give a score, not to rank the project, not to blindly praise it, not to invent customer or market data, to say honestly whether AI adds value, and to identify themselves accurately. The prompt text and screenshots are in the repository folder Ai-rewive/ and in Appendix A.

The prompt calls the reviewers "10 independent AI reviewers", but nine responses are recorded, and nine is the number shown on the portfolio.

Individual reviews (summarised)

Reviewer	Strength	Concern	Suggestion
GPT-5.6 Luna (OpenAI)	Addresses a clear operational problem and helps avoid preventable stockouts.	Core inventory logic is conventional; AI adds limited value if it only uses simple stock and sales calculations.	Get stock tracking and alerts reliable first; then use AI for demand trends, unusual sales patterns and explainable reorder suggestions.
Claude Sonnet 5.5 (Anthropic)	Stockouts are a real everyday pain, and the reorder-level workflow addresses it directly.	Manual sales and purchase entry is an adoption hurdle for busy owners; the "AI" is simple average-sales arithmetic.	Make data entry effortless (barcode scan, quick-tap sales, bulk import); use AI only where it adds value, such as forecasting with seasonality and lead times.
Gemini (Google)	Targets a genuine pain point for small kirana stores by automating stockout and reorder thresholds.	AI insights on top of a basic ledger feel like unnecessary feature bloat for small shops.	Drop the complex AI and make manual sales and purchase entry very fast.
Grok (xAI)	Targets a genuine pain for kirana-style shops that lose sales from late stockouts.	AI adds little beyond thresholds and averages that a spreadsheet already provides.	Focus the demo on offline-first mobile alerts and one-tap supplier reorders, not generic AI claims.
M365 Copilot (Microsoft)	Addresses a real inventory tracking problem by preventing stockouts through reorder alerts.	The AI insight appears basic and gives little beyond standard reorder-level calculations.	Add demand forecasting that considers sales trends and supplier lead times.
Meta AI (Meta)	Clear, relevant idea; automating reorder detection tackles manual tracking and missed sales.	The AI part feels forced; the Maggi example is a threshold check, not an intelligent insight.	Anchor AI to one concrete metric small retailers lack, such as predicted stockout date or dead-stock flagging.
Qwen3.7 (Alibaba Cloud)	Solves a tangible problem by automating manual stock tracking and reorder alerts.	AI insights may be overkill for simple linear reorder calculations and add complexity.	Perfect basic tracking and WhatsApp alerts before adding predictive AI.
Perplexity	Targets a real, frequent pain point by automating low-stock alerts and reorder triggers.	AI insights risk restating reorder-level logic unless fed richer signals (lead times, seasonality, supplier reliability).	Add simple demand forecasting (for example a moving average) and a one-tap reorder-to-supplier flow.
Siri (Apple)	Directly addresses the pain of manual stockouts for small, busy retailers.	AI insights currently state the obvious instead of predicting future demand.	Integrate with local supplier catalogs to automate the actual reordering, not just send alerts.

What the reviews agree on

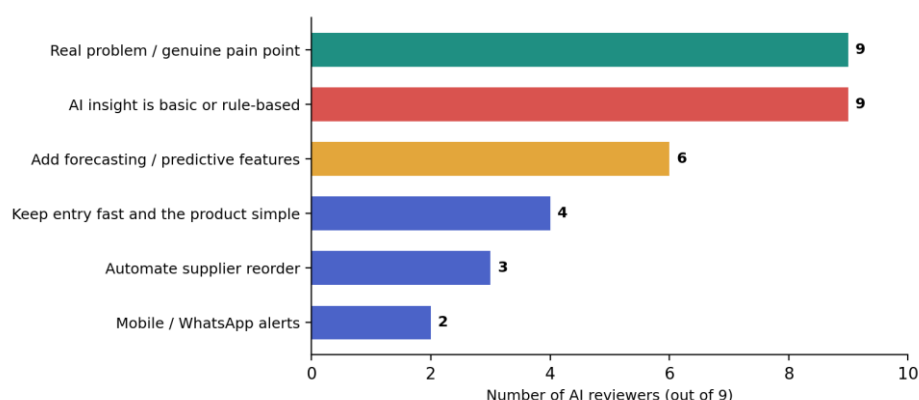


Figure 18: Themes in the nine reviews (grouping by the author of this report)

- **The problem is real (9 of 9).** Every reviewer said stockouts and manual tracking are a genuine pain for small retailers and that the reorder-level workflow addresses it.
- **The AI feature is the weak point (9 of 9).** Every reviewer said the AI insight is basic, obvious or no more than a threshold or average calculation. This is accurate: Section 4.3 shows that the prototype's insight is a rules-based calculation.
- **Forecasting would make AI worthwhile (6 of 9).** ChatGPT, Claude, Copilot, Perplexity, Meta and Siri asked for demand forecasting or prediction, using sales trends, lead times, seasonality or a predicted stockout date.
- **Simplicity and fast entry matter (4 of 9).** Claude, Gemini, Grok and Qwen stressed easy data entry and getting the basics right before adding AI.
- **Reordering should be actionable (3 of 9)** (one-tap supplier reorder, supplier catalogue integration), and two reviewers suggested mobile or WhatsApp alerts.

Rating summary

The reviewers were told not to give scores, so the star ratings were not part of the individual reviews. They come from a separate summary table (Figure A10) prepared afterwards from the nine responses, and the portfolio's 4.1 / 5 badge uses it. They are best read as a summary of the feedback, not as nine independent scores.

AI	Rating in summary table
Perplexity	4.5 / 5
Microsoft Copilot	4.5 / 5
Claude	4.0 / 5
GPT-5.6 Luna	4.0 / 5
Qwen	4.0 / 5
Grok	4.0 / 5
Siri	4.0 / 5
Meta AI	4.0 / 5
Gemini	4.0 / 5
Combined average (9 AIs)	4.1 / 5

How the feedback shapes ReStock

Feedback	Planned response
The AI insight is just threshold or average arithmetic	Keep the "Demo AI Insight" label. Move to a moving-average demand forecast, add supplier lead time and safety stock, and show the numbers behind each recommendation so it stays explainable (Phase 4 of the roadmap).
Manual data entry is an adoption hurdle	Add barcode scanning, quick-tap sales, bulk import and POS integration (Phase 3). The Sales window already pre-fills price and shows remaining stock to speed entry.
Alerts should lead to action	Turn a reorder request into a one-tap message to the supplier by WhatsApp or e-mail, and later link supplier catalogues (Phases 3 and 4).
Use channels shopkeepers already use	Add WhatsApp and e-mail notifications and keep working offline through local persistence.
Keep it simple	Keep the free Starter plan and the core workflow (record, detect, alert, reorder) as the first thing a new store sees.

Limits of this evidence. The reviews are opinions from AI assistants that saw only a short description. They show that the idea is understandable and that the AI feature needs work, but they are not market validation. Real validation needs the 30-day retailer pilot described in Section 3.8.

6. Results and Conclusion

Case Requirement Mapping

Case requirement	ReStock deliverable
Core customer problem	Stockout discovery and manual inventory monitoring
Problem and opportunity statement	Problem statement and product vision
Target customer segments	Small retailers, kirana stores, convenience stores, small pharmacies
Customer profile	Small Retail Shopkeeper persona
Value Proposition Canvas	Customer jobs, pains, gains, pain relievers and gain creators (3.2)
End-to-end customer journey	Eleven-stage journey from Discover to Improve (3.3)
Operational touchpoints	Dashboard, Products, Sales, Alerts, Reorders, Purchases, Suppliers
Supply-chain flow and responsibilities	Supplier to retailer to customer flow and responsibility table (3.5)
Revenue streams and cost structure	SaaS, analytics, multi-store, partnerships; development, infrastructure, support, acquisition (3.6)
Financial / performance measure	Stockout rate, alert response, reorder completion, stock cover, availability (4.4)

Case requirement	ReStock deliverable
SWOT and five or more risks	SWOT (3.9) and ten risks with mitigation (3.10)
Business model and go-to-market	Business Model Canvas, pilot retailers and distributor partnerships (3.6, 3.7)
90-day plan	30 / 60 / 90-day pilot-to-scale plan (3.8)
Demo	Interactive prototype on GitHub Pages

Findings

- The prototype covers the whole workflow: add product, record sales and purchases, calculate stock, detect low and out-of-stock products, alert, suggest a quantity, create a reorder, receive stock and keep history.
- The stock status rule and suggested-quantity formula are simple and transparent, so a shopkeeper can see why a product was flagged (for example Maggi: 8 in stock, level 10, 2 days of cover, order 20).
- Nine AI reviewers agreed that the problem is real and the workflow practical, and agreed that the AI feature is the weakest part. The roadmap therefore puts reliable tracking and easy data entry first and treats AI as a later, well-defined upgrade.

Limitations

- Prototype data is seeded, kept only in the browser, and there is no production database or multi-user login.
- The dashboard totals include placeholder demo figures, and the supplier contacts are demo data.
- AI Insights are rules-based and demonstrative; reorder calculations are simplified and ignore lead time, seasonality and minimum order quantities.
- No live supplier integration, payment or subscription system exists; the pricing plans are proposals.
- The 80-product store, 12 low-stock items and similar pilot figures describe the academic scenario, not measured results. No real retailer has been tested yet, and the AI reviews are not customer validation.

Future Scope

Phase	Possible additions
2: Production backend	REST API, authentication, store accounts, database, product CRUD, sales and purchase transactions
3: Retail integrations	Barcode scanning, POS integration, invoice import, supplier APIs, WhatsApp and e-mail notifications, mobile app
4: Intelligence	Demand forecasting, stockout prediction, supplier lead-time analysis, automatic reorder recommendations, seasonal analysis
5: Scale	Multi-store support, role-based access, central inventory, organisation accounts, advanced analytics, subscription billing

Conclusion

ReStock is built around one operational idea: help a retailer know what is running low before the customer finds out. The project shows this through a working prototype, a clear business model and a pilot plan, and it tests the idea against nine AI reviewers. Their feedback is consistent and useful: the problem is worth solving, but the value will come from dependable stock tracking, effortless data entry and actionable reordering, with AI added only where it does more than a simple calculation. The next step is a real 30-day retailer pilot to replace the demo data with measured results.

7. References

- ReStock repository and README, github.com/Yashhh710/restock-inventory-management.
- ReStock product portfolio and prototype, yashhh710.github.io/restock-inventory-management.
- A. Osterwalder and Y. Pigneur, Business Model Generation, John Wiley & Sons, 2010.
- A. Osterwalder, Y. Pigneur, G. Bernarda and A. Smith, Value Proposition Design, John Wiley & Sons, 2014.
- E. A. Silver, D. F. Pyke and D. J. Thomas, Inventory and Production Management in Supply Chains, 4th ed., CRC Press, 2017 (reorder point and safety stock concepts).
- S. Chopra and P. Meindl, Supply Chain Management: Strategy, Planning, and Operation, Pearson (supply-chain flows and responsibilities).
- MDN Web Docs, `Window.localStorage`, developer.mozilla.org (browser storage used by the prototype).
- GitHub Docs, GitHub Pages, docs.github.com (static hosting).
- AI reviewer responses from GPT-5.6 Luna, Claude Sonnet 5.5, Gemini, Grok, M365 Copilot, Meta AI, Qwen3.7, Perplexity and Siri, collected by the author (repository folder `Ai-revive/`).
- ITM Skills University, Go-to-Market & Customer Operations, Case Study: Retail Stockout Alert Service (problem statement).

Appendix A: AI Reviewer Screenshots

Screenshots of the nine AI responses and the rating summary, taken by the author. Prompts are shortened by each app's chat window; the full prompt is in Ai-rewive/prompt.txt. The Meta AI and Perplexity screenshots do not show the "Reviewed by" line, so only the product name is given for them.

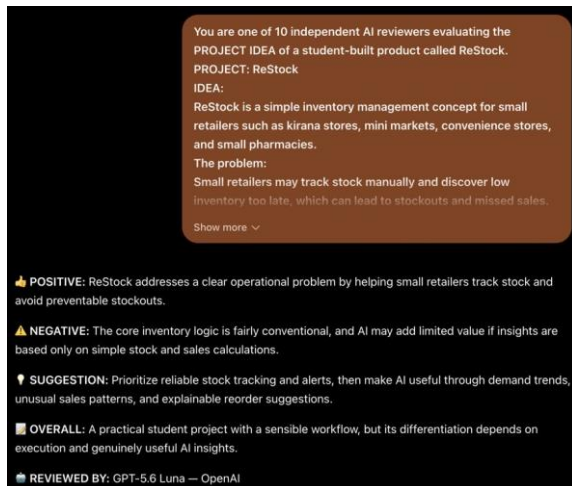


Figure A1: GPT-5.6 Luna (OpenAI)

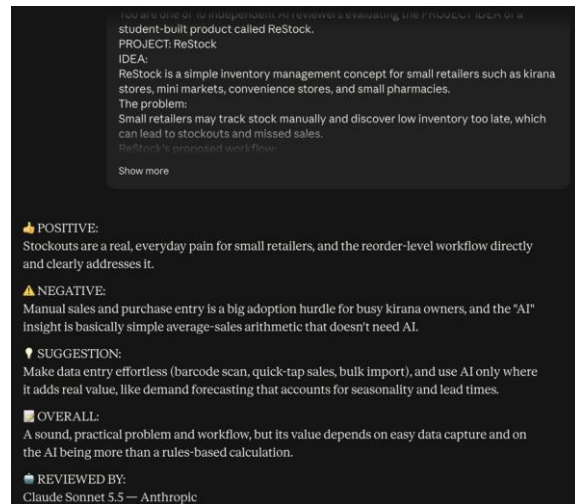


Figure A2: Claude Sonnet 5.5 (Anthropic)

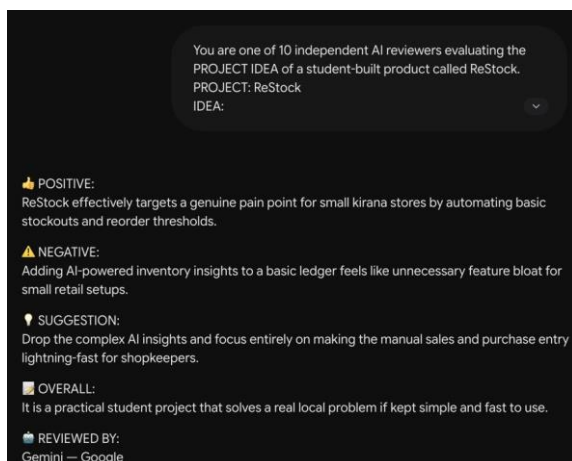


Figure A3: Gemini (Google)

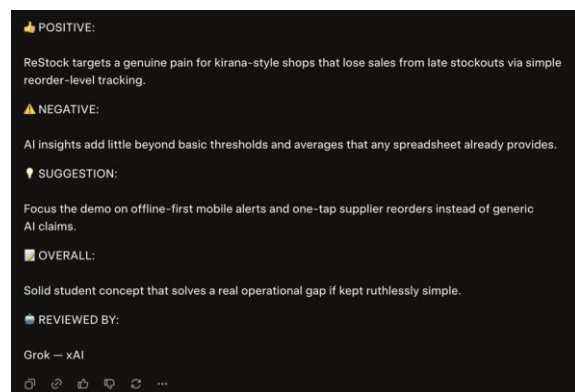


Figure A4: Grok (xAI)

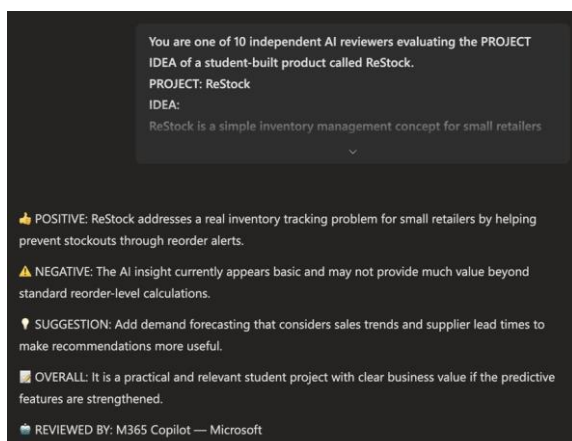


Figure A5: M365 Copilot (Microsoft)

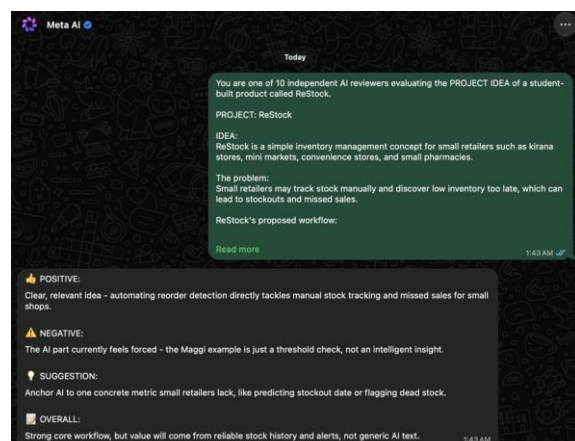


Figure A6: Meta AI

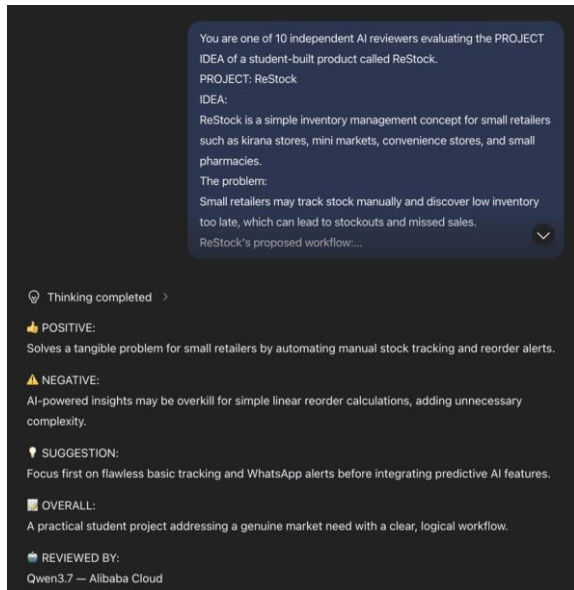


Figure A7: Qwen3.7 (Alibaba Cloud)

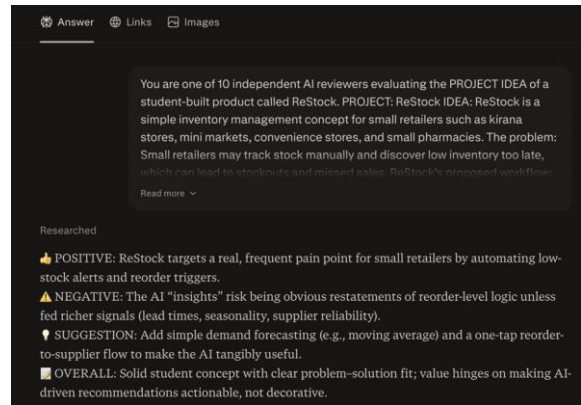


Figure A8: Perplexity

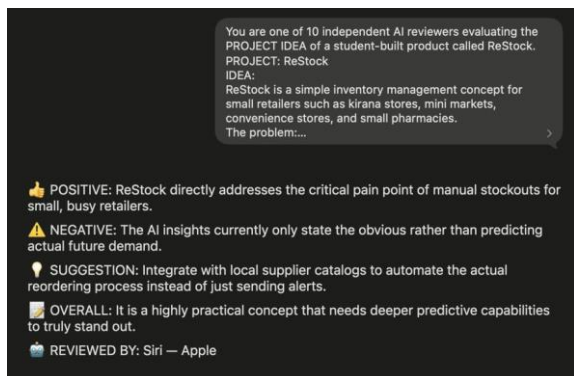


Figure A9: Siri (Apple)

★ ReStock — Individual AI Ratings		
AI	Rating	Main reason
Perplexity	★★★★½ 4.5/5	Clear problem-solution fit; suggested forecasting + actionable supplier reordering
Microsoft Copilot	★★★★½ 4.5/5	Practical project with clear business value; wants stronger predictive features
Claude	★★★★ 4.0/5	Strong real-world problem, but data entry and AI usefulness need improvement
GPT-5.6 Luna	★★★★ 4.0/5	Sensible workflow and real problem; differentiation depends on execution
Qwen	★★★★ 4.0/5	Genuine market need and logical workflow; recommends prioritizing basic tracking
Grok	★★★★ 4.0/5	Real operational problem; recommends keeping the product simple
Siri	★★★★ 4.0/5	Highly practical, but predictive capabilities need to go deeper
Meta AI	★★★★ 4.0/5	Strong core workflow; AI needs to provide more than threshold calculations
Gemini	★★★★ 4.0/5	Practical local problem; simplicity and speed are important
🏆 Combined AI Rating		
9-AI average: ★★★★★½ — 4.1 / 5		

Figure A10: Summary table of ratings